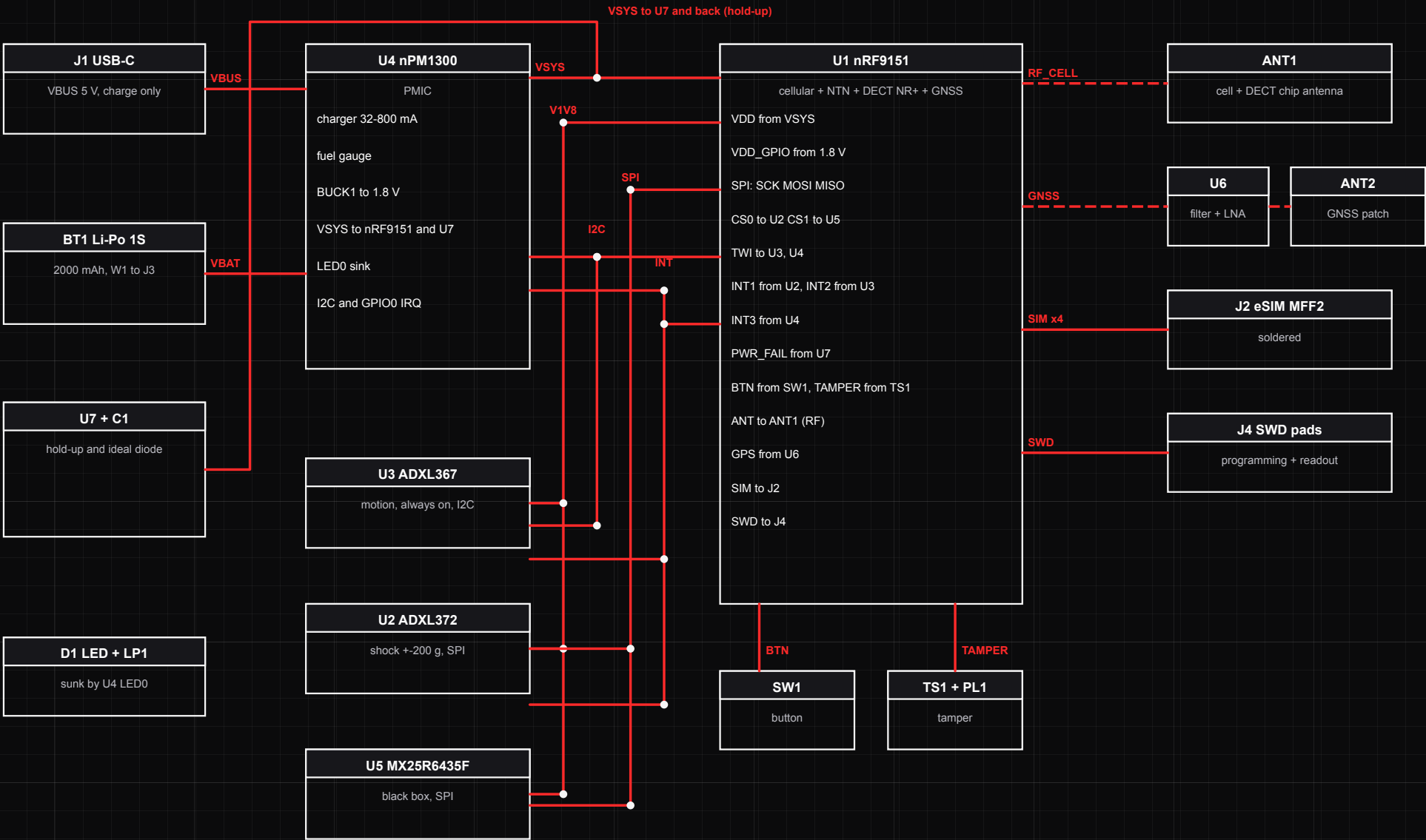


BLOCK DIAGRAM FROM hardware/netlist.json



Red solid: signals and supplies. Dashed: 50 ohm RF. One cable inside the device: W1 (BT1 to J3).

Hold-up: C1 0.5 F from 5 V to 3.3 V stores 3.53 J; the last-gasp budget needs 1.50 J (2.4x margin, assumptions in the netlist).

REV	DESCRIPTION	DATE
0	Hardware concept RL-MB-01	2026-10-05

RAILS (RANGES CHECKED)

RAIL	SOURCE	VOLTAGE	LOADS
VBUS	J1 USB-C	4.5-5.5 V	U4
VBAT	BT1 Li-Po 1S	3-4.2 V	U4
VSYS	U4 nPM1300 VSYS	3-5.5 V	U1 VDD, U7, D1
V1V8	U4 nPM1300 BUCK1	1.8 V	U1 GPIO, U2, U3, U5, U6, J4
VCAP	C1 hold-up capacitor	3.3-5 V	C1 via U7

BUSES

BUS	MEMBERS	SELECT / ADDRESS
SPI	U2 ADXL372, U5 flash	CS_IMPACT, CS_BLACKBOX
I2C	U3 ADXL367, U4 nPM1300	0x1D, 0x6B (to confirm)
RF	U1 to ANT1 ; ANT2 to U6 to U1	50 ohm, tuning to do

FIRMWARE POINTS TO FIX

- VDD_GPIO (V1V8) must rise more than 6 ms after VDD and ENABLE: sequence BUCK1 of the nPM1300.

- I2C addresses of the ADXL367 and the nPM1300 are to be confirmed on their data sheets before routing.

- nRF9151 GPIO assignment (CS0, CS1, INT1-3, BTN, TAMPER, PWRFAIL, COEX0) is fixed at routing.

- Black box readout: no USB data on this board. Read the flash through the programming pads (SWD or a UART bridge on them) after recovery, or upload it over LTE-M when a link exists.

COMMUNICATION LAYERS

- Cellular LTE-M / NB-IoT: normal reporting.
- Fleet relay over DECT NR+: the message hops through any passing fleet vehicle.
- Satellite NB-NTN: optional, off by default. Needs its own L-band antenna.

Priority 3 (theft, tamper, power loss) leaves first on whichever layer is up.

GENERAL NOTES

- Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).
- Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.
- Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.
- Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

ELECTRICAL ARCHITECTURE

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO. RL-EL-04	SCALE diagram, not to scale	REV 0	SHEET 4 / 5
DRAWN BY: M. ABAAQIL	DATE: 2026-10-05	MATERIAL: to be defined	